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VIR-CDI: Novel Voltage-Incremented Ramping Capacitive Deionization for Energy Efficient, Scalable, and Selective Physicochemical Nitrate Removal from Water

IARCO 2026 Research Proposal

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Research Problem

Nitrate contamination threatens drinking-water safety and aquatic ecosystems. The United States Environmental Protection Agency sets the maximum contaminant level for nitrate at 10 mg/L as nitrogen because excessive exposure can impair oxygen transport in infants [1]. Agricultural runoff, wastewater, and other nitrogen inputs can raise nitrate concentrations, while conventional nitrate control often depends on ion exchange, reverse osmosis, or biological denitrification. These approaches can be effective, but they may require chemical regeneration, high pressure, carbon dosing, or management of a concentrated residual. A low-pressure electrochemical process that selectively removes nitrate and can regenerate its electrodes could add a useful treatment barrier for water reuse and decentralized applications.

Capacitive deionization (CDI) removes ions by applying a potential across porous electrodes. Ions are stored electrostatically in the electrical double layer and released when the cell is discharged. CDI can remove nitrate in batch systems [2], but common constant-voltage operation does not actively manage the order in which ions enter the pores. Chloride and other abundant anions can occupy adsorption sites before nitrate is captured. Nitrate selectivity also depends on concentration, hydration, pore structure, membrane transport, and the operating voltage [3]. These dependencies make operating strategy a plausible route to selectivity, especially when new electrode materials would be expensive or difficult to manufacture.

Preliminary VIR CDI Results

My preliminary project tested voltage incremented ramping CDI, abbreviated VIR CDI, as an operational alternative to constant voltage. A 600 mL batch cell used 5 cm by 5 cm activated carbon cloth electrodes, stainless-steel current collectors, and a nylon separator. Six eight-minute voltage profiles were tested in triplicate, for 18 total runs. The control remained at 1.2 V.

Ramped profiles began at 0.4 V and increased toward 1.2 V with 2, 4, 8, 16, or 32 increments. The strongest result came from a two-stage profile that held 0.4 V during the first half of the adsorption period and then stepped to 1.2 V.

Across the synthetic nitrate-chloride tests, the maximum reported nitrate removal was 92.6%. VIR profiles produced nitrate-over-chloride selectivity values of 1.75 to 2.30, compared with approximately 1.0 for constant 1.2 V. Effluent chloride remained between 129 and 183 mg/L after VIR treatment, compared with 71.4 mg/L after constant-voltage treatment, indicating that VIR adsorbed less of the competing chloride. Maximum nitrate productivity was 12.8 mg/g/min for VIR and 10.1 mg/g/min for the control. The standard deviation in nitrate removal across triplicates was approximately 0.7 to 1.3 percentage points.

These results support further study but do not establish field performance. The experiment used a simplified batch matrix, short adsorption periods, and limited cycling. It did not test continuous flow, regeneration, natural organic matter, sulfate, hardness, or day-to-day wastewater variability. In addition, the preliminary energy values require recalculation from the raw voltage and current records before they can support a claim. The proposed research therefore treats the completed work as proof of concept and tests whether the VIR advantage persists in a real reuse-water matrix under repeated operation.

Existing Literature

CDI performance cannot be judged by salt removal alone. Hawks et al. recommend reporting water recovery, productivity, energy consumption, charge efficiency, and clearly defined test conditions so that systems can be compared objectively [4]. For selective nitrate removal, the same principle requires separate measurements of nitrate and competing ions rather than relying only on conductivity. Tang et al. demonstrated nitrate removal by batch CDI [2], while Tsai et al. showed that CDI and membrane CDI can favor nitrate over chloride and that selectivity changes with operating conditions [3]. Hawks et al. also achieved nitrate preference through ultramicroporous carbon [5]. Together, these studies establish that nitrate selectivity is possible, but much of the reported improvement depends on electrode or membrane design.

Voltage waveforms provide another control variable. Ramachandran et al. showed that sinusoidal operation can change CDI energy efficiency by matching the electrical input to cell dynamics [6]. Dykstra et al. compared constant-current and constant-voltage operation and demonstrated that charging protocol affects energy use [7]. These studies focused mainly on desalination efficiency rather than a monotonic voltage program designed to delay competitor-ion adsorption. Nitrate removal and recovery studies also show that regeneration and cycling determine whether adsorption performance can translate into a practical process [8]. Process models can connect cell measurements to hydraulic throughput and cost, but their conclusions depend on accurate current, volume, and cycle data [9].

The unresolved problem is therefore specific: it is unknown whether a simple two-stage 0.4-to-1.2 V profile can improve nitrate selectivity and energy use in continuous-flow secondary effluent without specialized electrodes. Real wastewater introduces dissolved organic matter, sulfate, phosphate, alkalinity, calcium, magnesium, and suspended material that can compete for charge, block pores, or change electrical resistance. Repeated adsorption and regeneration can also reduce capacity. A paired study using time-matched wastewater aliquots is needed to separate the effect of voltage profile from changes in feed chemistry.

Research Question and Hypotheses

The research question is: under continuous-flow treatment of secondary effluent from Orange County Sanitation District, how does a two-stage VIR profile of 0.4 V followed by 1.2 V compare with constant 1.2 V operation in nitrate removal, nitrate selectivity over competing anions, specific energy consumption, productivity, and stability during repeated adsorption and regeneration cycles?

H1 VIR will reduce specific energy consumption while maintaining nitrate removal within five percentage points of the constant-voltage control.

H2 VIR will increase the molar nitrate-over-chloride selectivity coefficient because the lower initial field will reduce early nonselective chloride adsorption.

H3 Real secondary effluent will lower absolute removal relative to synthetic water, but the relative VIR advantage will remain detectable after accounting for sampling day and electrode pair.

H4 A standardized adsorption and 0 V regeneration sequence will retain at least 90% of initial nitrate capacity after 50 cycles without measurable accumulation of nitrite or ammonium.

Methodology

Study Design and Source Water

The study will proceed in three stages. The exact OC San facility, process stream, sampling schedule, and onsite work conditions will be finalized through written site approval. The planned feed is secondary treated wastewater collected as refrigerated composite samples and analyzed within the approved holding time. Each composite will be measured for nitrate, nitrite, ammonium, chloride, sulfate, phosphate, conductivity, pH, temperature, alkalinity, hardness, calcium, magnesium, dissolved organic carbon, and turbidity. If native nitrate is too low for a precise treatment comparison, matched aliquots will be spiked to 10 and 25 mg/L as nitrogen while an unspiked arm is retained. These values will be described as experimental concentrations, not as OC San discharge limits.

Stage One Bridge Experiment

A bridge experiment will verify the laboratory ranking with corrected concentration labels and synchronized current logging. Three profiles will be tested in both a verified synthetic matrix and OC San water: constant 1.2 V, VIR50 with 0.4 V for half of the adsorption period followed by 1.2 V, and VIR25 with the step after one quarter of the period. Each condition will receive at least three independent cycles. Run order will be randomized, and matched electrode pairs will rotate among treatments. The profile that gives the best prespecified balance of nitrate removal, selectivity, energy, and productivity will advance.

Stage Two Continuous Flow Validation

A sealed flow-by CDI cell will use activated carbon cloth, stainless-steel current collectors, and a nonconductive separator. Electrode active mass and geometric area, spacer volume, hydraulic volume, tubing volume, and pressure drop will be documented. Three residence times will first be screened. The selected residence time will then remain fixed for the confirmatory comparison. On each sampling day, one composite sample will be divided among the selected VIR treatment, constant 1.2 V control, and a 0 V hydraulic blank. The split-sample design holds feed chemistry constant. Treatment order will be randomized, while sampling day and electrode pair will serve as blocks.

The staged sequence will first select the voltage profile, then fix the residence time, then estimate the real-matrix treatment effect, and finally measure durability. This order prevents exploratory choices from changing after the confirmatory comparison begins.

After adsorption, the cell will be short-circuited at 0 V and flushed with a fixed water volume until conductivity stabilizes. The concentrate will be collected to calculate nitrate recovery and regeneration-water demand. The target confirmatory dataset is five independent sampling days with three paired cycles per treatment per day, for 15 paired comparisons. The final sample size will be checked against the raw preliminary variance and a five-percentage-point minimum relevant difference before testing begins.

Measurements and Data Analysis

Performance Measures

Nitrate, nitrite, ammonium, chloride, sulfate, and phosphate will be measured in influent, treated water, and regeneration concentrate using validated laboratory methods. Conductivity, pH, temperature, flow, and pressure will be recorded for every cycle. Voltage and current will be logged at least once per second. Calibration standards will bracket expected concentrations. Blanks, duplicates, and matrix spikes will be included for at least 10% of samples. Every run will have a unique identifier linked to feed chemistry, electrode mass, active area, flow, voltage profile, and raw current trace. Regulatory comparisons will use mg/L as nitrogen; nitrate-ion values will be reported only in a separate labeled column.

Nitrate removal 100(Cin - Cout)/Cin, where Cin and Cout are nitrate concentrations in the influent and treated water.

Specific energy consumption the time integral of cell voltage multiplied by current, divided by treated-water volume. Gross energy will be reported; recovered energy will be reported only if it is directly measured.

Selectivity the ratio of molar nitrate adsorption to molar chloride adsorption, normalized by each ion's molar availability in the influent. The equation and handling of values near zero will be locked before analysis.

Productivity milligrams of nitrate-nitrogen removed per gram of active electrode per minute, accompanied by treated volume per unit electrode area and time.

Charge efficiency ionic charge removed divided by electronic charge supplied. Cycle retention will equal capacity in cycle n divided by capacity in cycle 1.

Statistical Analysis

The confirmatory analysis will use a paired mixed-effects model. Voltage profile, water matrix, and residence time will be fixed effects. Sampling day and electrode pair will be random blocking effects. A profile-by-matrix interaction will test whether the VIR effect changes in secondary effluent. Estimated treatment differences will be reported with 95% confidence intervals. Residuals will be checked for unequal variance and non-normality; a transformation or rank-based sensitivity analysis will be used if model assumptions remain unsatisfied. Nitrate removal and specific energy consumption will be coprimary outcomes. Selectivity, productivity, charge efficiency, nitrate recovery, and pressure drop will be secondary outcomes. This hierarchy limits conclusions drawn from multiple comparisons.

Durability Safety and Data Integrity

The selected VIR profile and constant-voltage control will each complete 50 adsorption-regeneration cycles. Removal capacity, energy, current response, pressure drop, and

nitrate recovery will be tracked. A persistent capacity loss greater than 10% will trigger inspection and standardized physical cleaning. If available, electrochemical impedance spectroscopy and cyclic voltammetry will compare electrodes before and after cycling. Nitrite and ammonium will be monitored for unintended nitrate conversion.

Secondary effluent will be handled under facility safety procedures using personal protective equipment, closed containers, spill control, and approved disinfection. Connections will remain dry and current-limited, and power will be disconnected before opening the cell. Concentrate will follow the approved disposal pathway. Raw files, scripts, calibration records, and data tables will be retained. No human or animal subjects are involved.

Research Topic and Significance

Most selective CDI research changes membrane or electrode chemistry. VIR CDI instead changes the voltage schedule while using commercially available activated carbon cloth. If the schedule improves nitrate selectivity and energy use, it could be added through power-control software rather than through a new electrode manufacturing process. The study also tests a mechanism suggested by the preliminary data: a lower initial field may limit rapid chloride uptake, leaving more pore capacity for nitrate when the voltage increases. Measurements of ion-specific adsorption and current response will determine whether that explanation is consistent with the observed performance.

Orange County Sanitation District is a relevant setting because it supplies more than 170 million gallons per day of treated wastewater to the Orange County Water District for the Groundwater Replenishment System [10]. The proposed work does not assume that OC San has a nitrate compliance problem and does not imply district endorsement. Its purpose is to expose VIR CDI to a complex reuse-water matrix and generate data that a utility could use to decide whether a larger pilot is justified. If onsite access is delayed, the same paired protocol will be conducted offsite with preserved OC San samples and described as a real-matrix bench validation.

Expected Outcomes and Decision Criteria

VIR will be considered successful only if it lowers specific energy consumption by at least 15% while keeping nitrate removal within five percentage points of constant 1.2 V. A useful selectivity result will require at least a 20% increase in the nitrate-over-chloride coefficient in real secondary effluent, with a confidence interval that excludes no improvement. The durability target is at least 90% capacity retention after 50 cycles and consistent nitrate recovery during regeneration. These thresholds are practical decision criteria rather than predicted results. Failure to meet them would show that the batch advantage does not survive real-matrix or continuous-flow constraints.

Limitations and Contingencies

A small flow cell cannot reproduce every full-scale hydraulic condition. Seasonal chemistry may not be represented by five sampling days, and spiking nitrate can alter ion ratios. These limits will be addressed by reporting complete feed chemistry, keeping an unspiked arm, blocking by day, and expressing flow by residence time and electrode area. Organic fouling will be evaluated through dissolved organic carbon, pressure drop, energy, and capacity trends. If current logging is incomplete, the affected cycle will be excluded from energy and charge-efficiency analysis rather than replaced with modeled values. If the 0 V blank shows nitrate loss, tubing and sample preservation will be investigated before electrosorption is interpreted.

Schedule and Deliverables

Weeks 1-2 will finalize access, reconcile records, lock equations, and approve safety and sampling plans. Weeks 3-4 will characterize water and complete bridge tests. Weeks 5-8 will conduct paired flow validation. Weeks 9-11 will complete durability and regeneration testing. Weeks 12-13 will finalize mass balances and statistics. Week 14 will produce a pilot report and scale-up recommendation.

The project will test whether voltage scheduling can provide selective nitrate control without specialized materials. A positive result will justify seasonal testing and stack-scale analysis. A negative result will define conditions where VIR loses its advantage. Either outcome will replace a synthetic batch observation with evidence for continuous reuse-water treatment.

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